

Nitin Singh

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Summary

Machine Learning Engineer at Aurora Innovation, specializing in on-device perception. I focus on the efficient design of large-scale machine learning models, aiming to scale them for high accuracy while leveraging quantization, TensorRT and custom CUDA kernels for optimized deployment.

Education

Carnegie Mellon University— Robotics Institute, Pittsburgh, PA

Master of Science in Computer Vision

Aug. '18 – Dec. '19

Indian Institute of Technology, Kanpur, Kanpur, India

Bachelors of Technology in Electrical Engineering,

Aug. '11 – Apr. '15

Work Experience

Aurora Innovation Inc, Mountain View, CA

Staff Software Engineer, Onboard Perception, 3D Object Detection

Jan. '21 – Ongoing.

- Led development of Aurora's on-device 3D object detection system using large vision models. Enabled autonomous vehicle to perceive and safely navigate complex environments.
- Built sensor fusion architectures integrating data from 5 LiDARs and 12 cameras. Leveraged calibration and time synchronization to achieve high precision and recall across diverse environmental and traffic scenarios, improving operational safety.
- Modernized the perception stack by designing transformer-based LiDAR encoders to replace dense convolutions. Increased model efficiency and extended detection range, enhancing autonomous vehicles safety.
- Designed and implemented custom CUDA kernels for LiDAR-camera fusion, scaling the system to process 12 camera images every 100 ms. Improved pedestrian and cyclist recall by 20% at long range.
- Enabled Int8 quantization of the model, reducing latency and broadening hardware support to scale detection model.
- Owned vehicle detection improvements to unblock commercial operations in challenging conditions, including night driving, sun glare, and highway merging.

Uber, Advanced Technologies Group(ATG), San Francisco, CA

Autonomy Engineer II, Perception-Prediction Group

Feb. '20 – Jan. 21

- Implemented joint LiDAR-centric and camera-centric detection architecture instead of two separate inference streams. This enhanced sensor fusion and reduced compute required for inference while maintaining performance.

Samsung Research Institute, Bangalore, India

Senior Software Engineer with Camera Solutions Group

May. '15 – Aug. '18

- Developed the Bokeh effect for Samsung camera using dual-focus images and a CNN-based segmentation model. Integrated data from both dual-focus and segmentation through graph cuts to achieve high segmentation accuracy at edges, enabling commercialization across all devices worldwide.
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Academic Projects

Trajectory Prediction for Self-Driving Cars

Guide: Prof. Jeff Schnieder

Jan. '19 – Dec. '19

Used prior motion cues, scene features and interactions between agents to predict future trajectory of actors. **Report.**

Publications

- Uncertainty-aware Short-term Motion Prediction of Traffic Actors for Autonomous Driving, IEEE Winter Conference on Applications of Computer Vision (WACV '20) **Paper, Open Access, CVF.**
- Depth Aware Portrait Segmentation using Dual Focus Images. IEEE International Conference on Multimedia and Expo (ICME), 2018. **IEEE.**
- Saliency Prediction with Deep Multiscale Residual Network. Computer Vision and Image Processing 2018. **Paper**

Patents

- Method and image capturing device for generating high resolution image of region of interest. **Reference**
 - Electronic device and method for automatic human segmentation in image: **Reference**
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Courses & Technical Skills

Grad Courses: Computer Vision, Machine Learning, Visual Learning and Recognition, Geometry in Vision, Math Fundamental for Robotics, Simultaneous Localization and Mapping

Programming: Python, C++

Frameworks : Pytorch, Tensorflow